RB-205: MySQL Primary Failover

Field	Value
Service	billing-db (MySQL 8.0, Orchestrator cluster alias billing-db-prod)
Severity	SEV-1
Owner	Billing Engineering team
Last reviewed	2026-08-27

1. Symptoms

- Alert MySQLPrimaryDown fires when mysql_up == 0 on the primary for 1 minute.
- Orchestrator reports the cluster as DeadMaster on its web UI (orchestrator.internal, port 3000).
- billing-api logs ERROR 2003 (HY000): Can't connect to MySQL server and the hourly invoice job stalls.

2. Cluster topology

- mysql-billing-01: primary in normal operation.
- mysql-billing-02: semi-sync replica and the preferred promotion candidate (promotion_rule=prefer).
- mysql-billing-03: async replica for reporting, never promoted (promotion_rule=must_not).

Applications connect only through ProxySQL on port 6033. Writes go to hostgroup 10, reads to hostgroup 20. The ProxySQL admin interface listens on port 6032.

3. Pre-checks

3.1 Confirm the primary is really down

```
orchestrator-client -c topology -i mysql-billing-01:3306
mysqladmin -h mysql-billing-01 -u orc_check ping
```

If the primary answers from the bastion but not from the ProxySQL hosts, this is a network partition, not a failover case: stop here and page Edge Networking (PagerDuty: edge-net).

3.2 Check replication lag on the candidate

Run SHOW REPLICA STATUS\G on mysql-billing-02. Promotion is allowed only if Seconds_Behind_Master is at most 30 seconds and both Replica_IO_Running and Replica_SQL_Running are Yes. If lag is above 30 seconds, wait for it to drain; do not promote mysql-billing-03 instead.

3.3 Check semi-sync status

```
SHOW GLOBAL STATUS LIKE 'Rpl_semi_sync%';
```

On mysql-billing-02, Rpl_semi_sync_replica_status must be ON. The source waits for 1 replica acknowledgement with a timeout of 10000 ms. If semi-sync had fallen back to async (Rpl_semi_sync_source_no_tx above 0 on the old primary), record gtid_executed from both hosts before going further.

4. Failover procedure

4.1 Fence the old primary

If mysql-billing-01 is still reachable, make it refuse writes:

```
SET GLOBAL super_read_only = ON;
SET GLOBAL read_only = ON;
```

If it is unreachable, power it off through its management controller: ipmitool -H mysql-billing-01-ipmi -U ops chassis power off.

4.2 Promote the replica with Orchestrator

If the old primary is alive and fenced, use the graceful takeover; if it is dead, use recover:

```
orchestrator-client -c graceful-master-takeover -a billing-db-prod -d mysql-billing-02:3306
orchestrator-client -c recover -i mysql-billing-01:3306
```

Orchestrator sets read_only=OFF on mysql-billing-02 and repoints mysql-billing-03 to it.

4.3 Repoint ProxySQL

The mysql_replication_hostgroups monitor moves the writer automatically within 60 seconds. If it has not, on the admin interface (port 6032):

```
UPDATE mysql_servers SET hostgroup_id=10 WHERE hostname='mysql-billing-02';
DELETE FROM mysql_servers WHERE hostname='mysql-billing-01' AND hostgroup_id=10;
LOAD MYSQL SERVERS TO RUNTIME;
SAVE MYSQL SERVERS TO DISK;
```

Warning: do not set read_only=OFF on two billing-db hosts at the same time, and never promote mysql-billing-03. Two writable hosts cause split-brain and duplicate invoice numbers.

5. Verification

- orchestrator-client -c topology -i mysql-billing-02:3306 shows mysql-billing-02 as primary with mysql-billing-03 replicating.
- SELECT hostgroup_id, hostname, status FROM runtime_mysql_servers; shows only mysql-billing-02 in hostgroup 10, status ONLINE.
- billing-api 5xx rate is back below 0.5% within 5 minutes and the invoice job resumes.

6. Rollback

Fail back only in a maintenance window, never in the first 24 hours. Rebuild mysql-billing-01 as a replica of mysql-billing-02 with orchestrator-client -c relocate -i mysql-billing-01:3306 -d mysql-billing-02:3306, wait until Seconds_Behind_Master is 0, then run a graceful-master-takeover back to mysql-billing-01. If the old primary accepted writes after fencing (errant GTIDs), re-clone it with CLONE INSTANCE instead of reattaching it.

7. Escalation

Level	Who	Contact
L1	On-call SRE	#ops-oncall
L2	Billing Engineering on-call	PagerDuty: billing-eng
L3	DBA lead (data loss, errant GTIDs, failed promotion)	PagerDuty: billing-eng, escalation policy "dba-lead"

See also: RB-204 PostgreSQL Primary Failover.